

# Electrical Machine- II (BT EE403A)



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## Unit-III

- 1. Induction Generator**
- 2. Self –excited Generator**

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# Induction Generator

Induction machines are sometimes used as a generator. These are known as **induction generators** or **asynchronous generators**. So under what conditions will an induction machine will behave as an induction generator?

An induction machine will behave as an induction generator when:

- Slip becomes negative due to this the rotor current and rotor emf attains negative value.
- The prime mover torque becomes opposite to electric torque.

Now let us discuss how we can achieve these conditions. Suppose that an induction machine is coupled with the prime mover whose speed can be controlled. If the speed of the prime mover is increased such that the slip becomes negative (i.e. speed of the prime mover becomes greater than the synchronous speed).

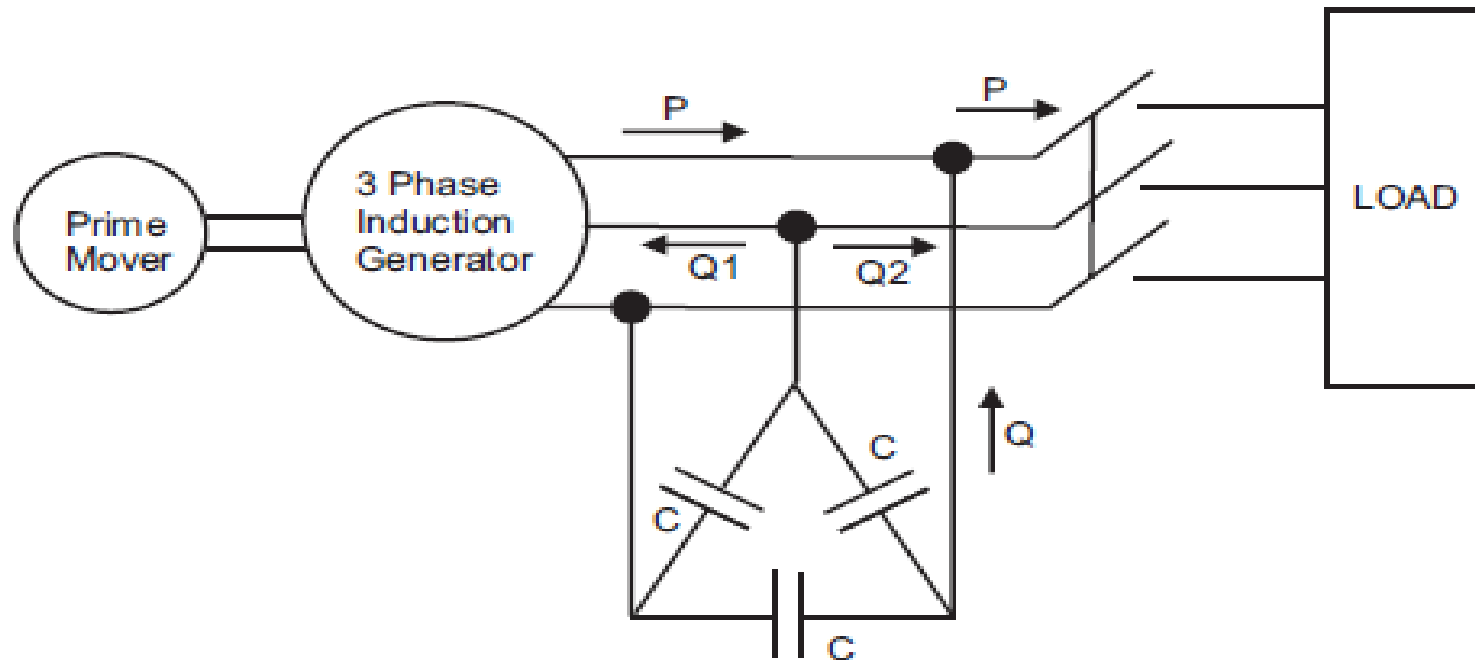
➤ Due to this, all the conditions that we have mentioned above will become fulfilled and the machine will behave like an induction generator. Now if the speed of the prime mover is further increased such that it exceeds the negative maximum value of the torque produced then the generating efficiency of the generator vanishes. Clearly, the speed of the induction generator during the whole operation is not synchronous, therefore the induction generator is also called a [synchronous generator](#).

➤ An induction generator is not a self-excited machine. Therefore in order to develop the [rotating magnetic field](#), it requires magnetizing current and reactive power. The induction generator obtains its magnetizing current and reactive power from the various sources like the supply mains or it may be another synchronous generator.

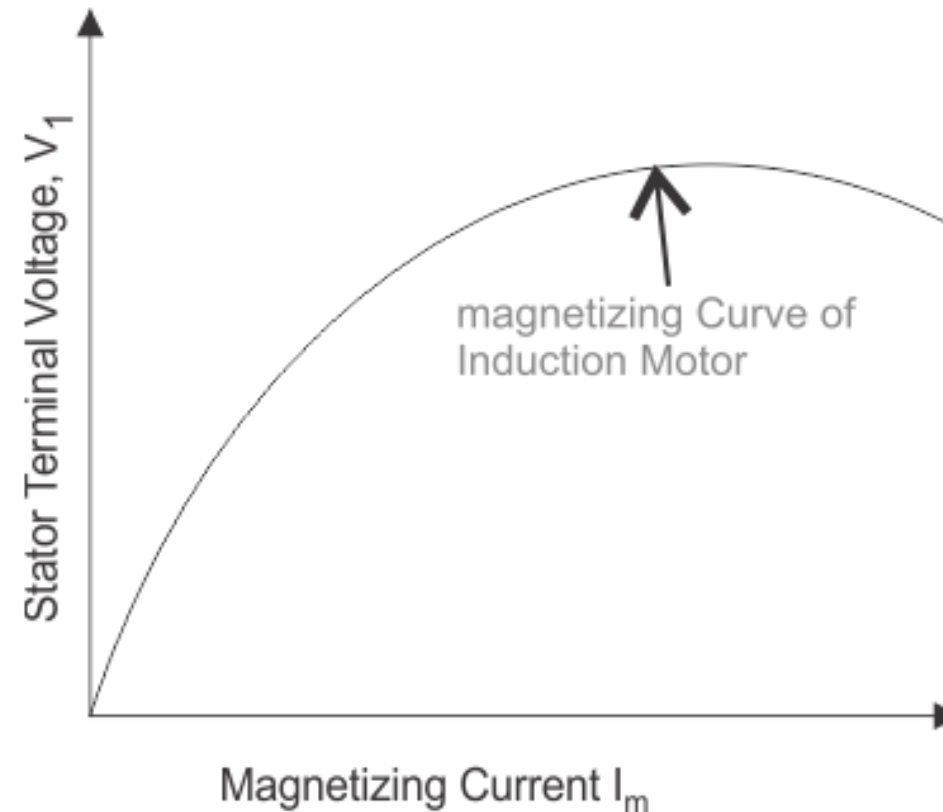
An induction generator can't work in isolation because it continuously requires reactive power from the supply system. However, we can have a self-excited or isolated induction generation if we use a [capacitor bank](#) for reactive power supply instead of an AC supply system. We'll now discuss **isolated induction generators** in detail.

# Isolated Induction Generator

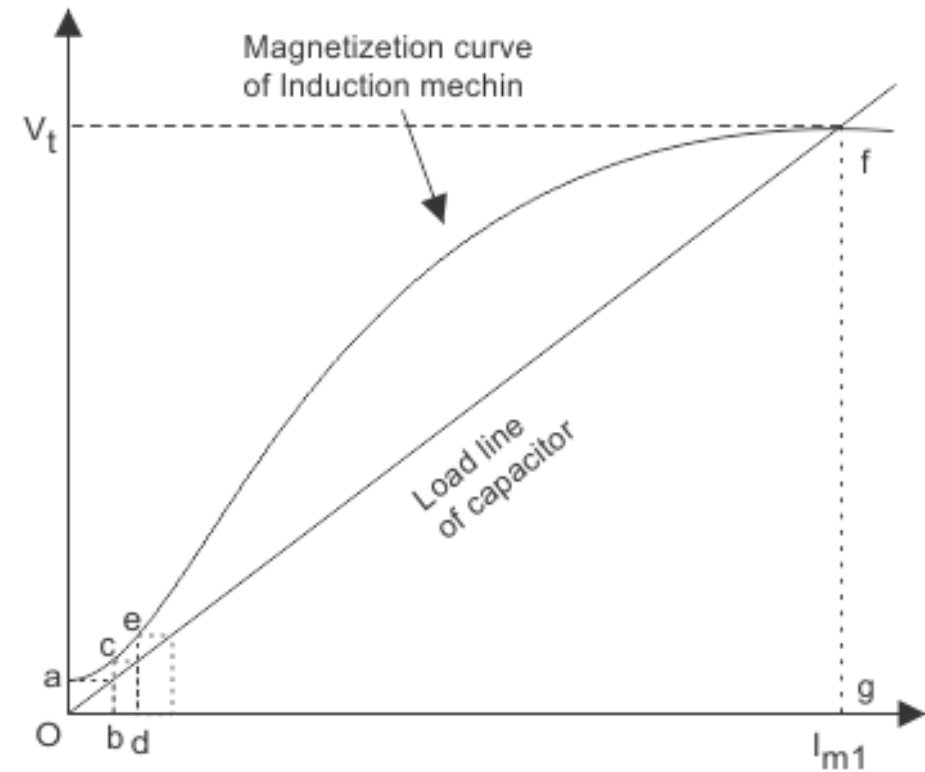
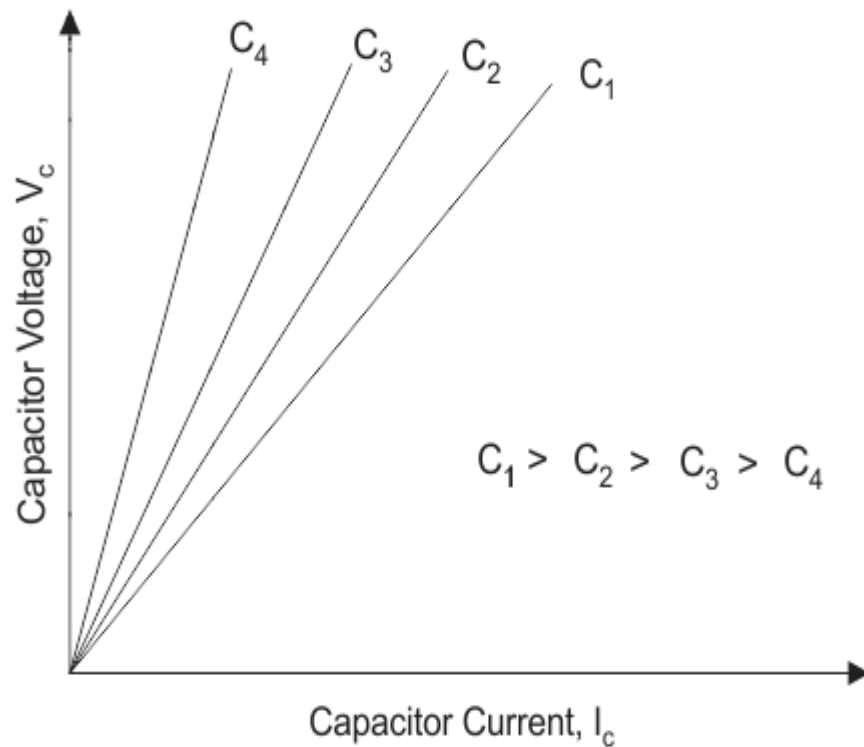
This type of generator is also known as a [self excited generator](#). Now why it is called self-excited? It is because it uses a [capacitor bank](#) which is connected across its stator terminals as shown in the diagram given below.



The function of the capacitor bank is to provide the lagging reactive power to the induction generator as well as load. So mathematically we can write total reactive power provided by the [capacitor bank](#) is equals to the summation of the reactive power consumed by the induction generator as well as the load.



There is generation of small terminal voltage  $oa$  (as in figure given below) across the stator terminal due the residual magnetism when the rotor of the induction machine runs at the required speed. Due to this [voltage](#)  $oa$  the capacitor current  $ob$  is produced. The [current](#)  $bc$  sends current  $od$  which generates the voltage  $de$ . The cumulative process of voltage generation continues till the saturation curve of the induction generator cuts the [capacitor](#) load line at some point. This point is marked as  $f$  in the given curve.



# Application of Induction Generator

Let us discuss **application of induction generator**: We have two types of **induction generator** let us discuss the application of each type of generator separately: Externally excited generators are widely used for regenerative braking of hoists driven by the [three phase induction motors](#).

Self-excited generators are used in the wind mills. Thus this type of generator helps in converting the unconventional sources of energy into [electrical energy](#). Now let us discuss some disadvantages of externally excited generator:

- The efficiency of the externally excited generator is not so good.
- We cannot use externally excited generator at lagging [power factor](#) which major drawback of this type of generator.
- The amount of reactive power used to run these types of generator required is quite large.



## **Advantages of Induction Generators**

- It has robust construction requiring less maintenance
- Relatively cheaper
- Small size per kW output power (i.e. high energy density)
- It runs in parallel without hunting
- No synchronization to the supply line is required like a [synchronous generator](#)

## **Disadvantages of Induction Generators**

- It cannot generate reactive volt amperes. It requires reactive volt amperes from the supply line to furnish its excitation.